**Daniyal Ali Hameedi**, **Nationality: American and British**
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## Technical Skills

- **Programming Languages:** Python, C++, Rust, C#, MATLAB, Lean4, Mathematica, LaTeX.
- **Core Competencies:** Quantitative Modeling, Stochastic Inference, Non-linear Filtering, Monte Carlo Simulations, CUDA GPU Acceleration, Machine Learning, Representation Theory, Conformal Field Theory, Statistical Mechanics.
- **Research Interests:** Algorithmic Information Theory, Algorithmic Thermodynamics, Agent Foundations, Multi-Agent Systems, Statistical Mechanics of Learning, Mathematical Foundations of Intelligence.

## Research & Professional Experience

**Highforest Capital Ltd** London, UK
*Quantitative Researcher / Developer* 02/2021 – 11/2023, 01/2026 – Present
- Built CUDA-accelerated Monte Carlo simulations for Markov-Switching Multifractal (MSM) models and devised an exact symmetry-reduced filtering algorithm, reducing multivariate state-inference complexity from exponential to polynomial.

**Independent Research** London, UK
*Independent Researcher* 01/2026 – Present
**Ongoing projects:**
- **Non-linear Filtering / Inference Theory:** Investigating exactly solvable nonlinear filtering problems using representation-theoretic methods, including applications of vertex operator algebras to infinite-dimensional solutions of the Zakai equation.
- **Multi-Agent Dynamics / AI Safety:** Developing statistical-mechanical models of interacting reinforcement-learning agents to investigate the emergence and dynamical signatures of collective misalignment.

**King's College London** London, UK
*Research Associate (Part-Time)* 06/2025 – 12/2025
- Investigated Renormalization Group Flows between Yang-Lee and Imaginary Field Ising models using the Wetterich equation to confirm CFT predictions, winning KCL's self-sourced research award.

**Xenophon Systems** London, UK
*Co-Founder & CEO* 12/2022 – 12/2023
- Prototyped a prediction-market mechanism design for eliciting truthful signals about intelligence outcomes in high-leverage low-trust environments without leaking sensitive underlying information.

**London Institute for Mathematical Sciences (LIMS)** London, UK
*Undergraduate Research Assistant (Self-Directed)* 05/2023 – 09/2023
- Conducted research on machine learning applications in mathematics and string theory. Developed the `extdivga` machine learning library in Rust for generating optimal representations for exterior derivatives.

## Education

**King's College London** London, UK
*MSci in Physics and Theoretical Physics* 2021 – 2025
- **Grade:** First Class Honors (4.0 GPA equiv.)
- **Thesis:** *Regular and Logarithmic Conformal Invariance in Polymer Gelation (81%)*
- **Academic Achievements:** Supersymmetry and Conformal Field Theory (95%), Quantum Field Theory (89%).

**Westminster School** London, UK
*A-Levels & GCSEs* 2015 – 2020
- **A-Levels (2020):** Mathematics (A), Physics (A), Economics (B), Further Mathematics (C).

## Pre-prints & Publications

**D. A. Hameedi**, *From Exponential to Polynomial: An Exact Filter for High-Dimensional MSM Models*, 2026.
Solo paper. Developed an exact symmetry-reduced filtering algorithm that reduces the computational complexity of state-inference in multivariate MSM models from exponential to polynomial.
- Pre-print: arXiv:2608.22864. URL: https://arxiv.org/abs/2608.22864.
- Under review at the Journal of Computational Economics.

## Conferences, Competitions & Awards

- **Iliad Intensive Course (10/2026):** Selected participant in a four-week intensive research program exploring mathematically rigorous approaches to AI safety, including Markovian Decision Theory, Solomonoff Induction, AIXI, Multi-Agent Dynamics, and Agent Foundations.
- **IOP International Physicists' Tournament (2022 – 2023, 2023 – 2024):** Represented the UK in the International Finals at ETH Zurich in 23/24 Tournament. Placed 2nd at UK National Team Selection in 22/23 Tournament.
- **KCL Theoretical Particle Physics & Cosmology Group (2024):** Invited to meetings of Prof. Eugene Lim's research group.
- **String DANGER (07/2023, 07/2024):** 2-Day Conferences on ML applications for research in String Theory, Data Science, Number Theory and Geometry.
- **Oxford University ML Summer School (07/2023):** Invited to 5-day bootcamp applying machine learning to string theory and geometry aimed at graduate students and post-docs.
- **Purple and Gold Scholar (2021):** University of Washington, Seattle (awarded highest scholarship level).